



COLOR THEORY BASICS

Color theory basics

Introduction

We can not separate painting from color theory. Even if we were to paint something in black & white, there are underlying factors at play that make the finished painting appealing, like **value**. In this tutorial we will go through color theory basics, so we can use colors in a way, that'll make our models more appealing.



WHY should I care?

Because we are always using colors. Using them in a way to make the finished painting appear more harmonious (or disharmonious, if we want) can elevate our miniatures even further. Some aspects of color theory are inherently subjective, but there are many objective measures that can be used universally.

What you will learn

- Basic color theory terms and their usage
- How colors interact
- Choosing colors for your miniature
- Color mixing (pigment theory)

Color theory basics

Basic terms

Given the subjective nature of color, **there isn't just one universal color theory**. In fact, there are so many concepts and theories that a person could study color theory for many years and still have gaps in the knowledge.

However, we can look at the most useful concepts for us, hobbyists.

Value refers to how bright is the color. Higher value means lighter color and lower value means darker color.



low value



high value

Saturation refers to how pure and intense the color appears. More saturated colors will appear more vivid, while desaturated colors are not as „juicy“.



low saturation



high saturation

Temperature refers to warmth or coolness of a color. In general, yellows, oranges and reds are warm colors and greens, blues and some purples are considered cool.

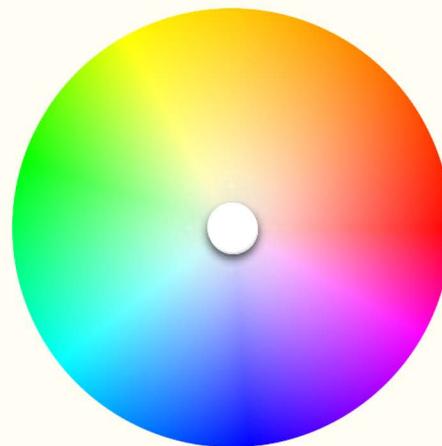


cool



warm

Finally, we have **hues**. This word describes a color we would think of normally, like yellow, green, orange, red or blue. This refers to pure colors, at their 100% saturation.



Pure colors can be found on the edge of the color wheel.

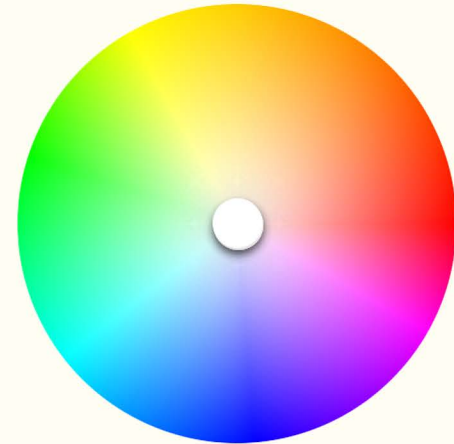
Color theory basics

Basic terms

Value is the most important metric. That's because even if you turn your miniautre black and white (without any saturation/temperature), it'll work. Even in black and white we can easily see which parts are exposed (higher value) and which are in shades (lower value). Recess shading and edge highlights really help in this way.



If your painting works in black and white, it will most likely work in color. However, we can use the color wheel to see how the colors interact when we think about their integration.



This is a color wheel is from [adobe.color](https://www.adobe.com/color) (great resource for artists). It should be one of the more accurate color wheels out there, because of the right spacing between the colors.

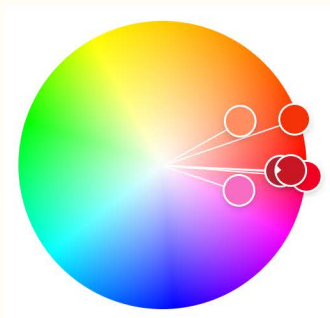
When selecting our color palette for the project, we can use **color harmonies** that might fit it. The most common color harmonies are:

- Monochromatic
- Analogous
- Complementary
- Triadic

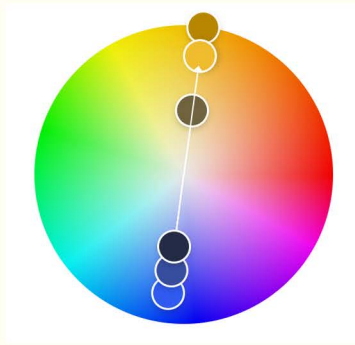
Color theory basics

Color harmonies

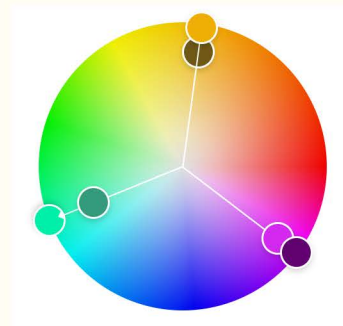
Here are some example of the aforementioned color harmonies:



Analogous color scheme consists of colors that are close to each other on the color wheel. The Scarlet Witch miniature (above) represents analogous color scheme, that leans into reds and magentas.



Complementary color harmony/scheme consists of two colors that are on the opposite sides of the color wheel. The hero in the golden armor has 2 opposing colors that make up majority of the miniature - yellow and blue.



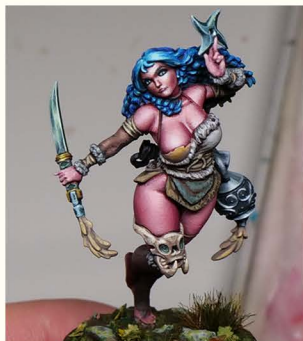
Although this incubi miniature is not finished, it represents a **Triadic** color scheme. We usually recognize primary color triad (red, blue, yellow) and secondary color triad (green, orange, purple), which was the one used in this case. The green and the purple are both present on the miniature and the orange is just sketched on the base.

Although they are very helpful, color harmonies should not be followed above everything else. They can help you make decisions about the most important parts of the model, but you still have to use other colors to paint all the accessories etc.

Color theory basics

Practical application

While color harmonies create aesthetically pleasing results, it's not obvious from them that you should also include **neutral colors**. These are colors that are not strong in saturation and usually don't capture viewers attention too much.



In this example we have just 2 strong elements that are highly saturated - the hair and the skin. **The rest of the miniature is more neutral and therefore doesn't take away the attention.** In general, we can choose more vivid colors for the parts we wanna highlight, while keeping the rest more muted.

This touches on an important subject - **contrast**. Everything we paint on our models creates more or less contrast - or difference. As mentioned on page 3, **the most important contrast is the one between values**. If we have enough contrast to differentiate lights and shadows, we are already doing pretty good. The above example points out **contrast in saturation**. There is also **contrast in temperature** and **contrast in color**.



Even though the crocodile is the focal point of the piece, notice the usage of the green/mint stone faces. This represents usage of **contrast in color**. The stone faces could be left grey, but by using green instead, they stand out a little bit.

Corona Girl represents **contrast in temperature**. Notice how the backdrop is colder than the center of the miniature - that makes the red armor and skintones stand out even further. It's also interesting to point out that **despite the fact that magenta would be warmer than blue/cyan, contextually it's colder than the red**.



Color theory basics

Practical application

When choosing a color scheme for your miniature, we recognize 2 main methods:

- Use the color harmonies to choose hues of the most important elements
- Pick one main color for the most important part and use more neutral ones for the rest

There is another alternative, which is just copying the box art or another artwork. You could also start with the colors you want and selectively add what you think will create more contrast (whether it's in color, saturation or temperature). Given that many miniatures have plenty of details, you'll most likely use a combination of these.



Sometimes you don't have to limit yourself in terms of color choice. Proper contrast in value (lightest colors near the face) and contrast in saturation (vivid red hair) will point viewer's attention to the right direction. There are also partial contrasts used - notice how the green grass and magenta shades in skintones make the mini stand out.

Pigments and color mixing

Until this point, we have only talked about how color theory presents itself on the finished miniature. Another important discipline is color mixing.



In the classic system color theory, red, blue and yellow are the 3 primary colors. Nowadays we are using cyan, magenta and yellow instead, because we can mix broader range of color (although imperfect, even printers use CMYK inks). Notice, how the colors get less saturated when we mix them. By mixing blue and red we got a desaturated purple, yellow and red gave us dull orange and blue with yellow got us vomit green. **Compare it to pre-mixed versions** of these colors below.



Color theory basics

Pigments and color mixing

By **mixing paints we usually lose saturation**. The feature (or bug) of using pigments is that the more variety of pigment you add, the more muddy is the result. However, **paints that use variety of different pigments often provide better coverage**. We can mitigate the loss of saturation by using paints that have only a few or just 1 pigment, like Kimera Colors or artist brands.



In this case we are using Phtalo Blue, Magenta and Cold Yellow. The resulting mixes are much purer and you can get broader range of colors. By mixing all of your primaries to a certain degree, you can also mix up browns and skintones.

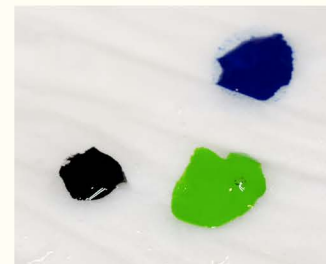
Aside from creating more hues, you can make individual color lighter or darker.



On the left we see a green paint we wanna make lighter. We can add lighter paints to do that - using just white will make the result lighter and desaturated (2). Using pale yellow will do the same thing, but it'll retain more saturation (1). Adding pale blue will make the result colder and lighter (3).

Although the differences might not be obvious at first, **it's a good practice to add other light colors rather than just white, to keep the result more saturated**. Mix (4) has even more yellow in it and the result is significantly more saturated.

You can do the same when darkening the color. For example, using blue (5) instead of black (6) will retain more saturation in the final mix.



Color theory basics

Pigments and color mixing

Sometimes it's desirable to mix in black/white/grey to make the resulting mix more muted. We don't wanna make everything stand out. **If everything stands out, nothing really stands out.** That's why we use different kinds of contrasts to bring attention to the spots we wanna highlight.



Picking the right paints

There is no thing such as the best paint. Different pigments and different products will behave in a different way. There is no need to fully stock up on artist paint ranges, although it is a good idea to have some.

It's also a good idea to have your workhorse paints (think citadel, AK interactive, Vallejo...) which have good coverage and we can achieve amazing results with them. We can also choose purer paints from those artist brands to enhance our results. Kimera Kolors, Liquitex Inks or Golden Acrylics are all useful in this regard.

Before committing to any particular paint brand, I highly recommend testing between 10 - 15 different colors to see if you like them.



Color theory basics

Final note and resources

Color theory can be as broad of a subject as you want it to be. This guide should provide you with the fundamentals, so you can start painting without second guessing your color mixing and color choice.

Given the subjective nature of colors, keep in mind that this is one of the „less exact“ aspects of our hobby and you might bend the color theory rules to create something specific, something you like. That is fantastic and don't let anyone stop you!

If you want a quick refresher on the stuff mentioned in this guide, I went over very similar lessons in one of my videos: „Choose Color Schemes For Warhammer Minis like a PRO!



Another resource I recommend is the color wheel from Adobe. Go to color.adobe.com to play with different color scheme for yourself!

Lastly, if you are a true color nerd, Peter Donahue is an amazing educator on TikTok and Instagram.

